

深圳市嘉佰达电子科技有限公司

SHENZHEN JIABAIDA ELECTRONICS TECHNOLOGY.CO.,LTD

产 品 规 格 书

Product specification

客户名 (CUSTOMER) :

产品名 (SAMPLE NAME) :

17-24 串 50-200A 软件板

产品型号 (MODEL NAME) :

JBD-SP24S004 V1.1 铁锂系列

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1.产品简介(Product introduction)

●JBD-SP24S004 是专门针对 17-24 串储能锂电池包而设计的软件保护板方案，该产品采用 0₂ 前端采集芯片+MCU 架构，部分参数可以根据客户需求，通过上位机灵活调整。

●JBD-SP24S004 is a software protection board scheme specially designed for 17~24 strings of lithium battery packs. The product adopts architecture of front-end acquisition chip + MCU, and some parameters can be flexibly adjusted through the host computer according to customer needs.

2.功能配置(Configuration)

功能(Function)	配置(Configuration)	功能(Function)	配置(Configuration)
支持串数 (Number of strings supported)	17~24S	485 通讯 (隔离) 485 communication (isolated)	选配 (Optional)
持续电流 (Continuous current)	50-200A	UART 接口 (隔离) UART interface (isolated)	不支持 (Not supported)
NTC 数量 (Number of NTC)	1 路内置，四路外置 (1 built-in, 4 external)	CAN 通讯 (CAN communication)	选配 (Optional)
均衡功能 (Balance Function)	被动均衡 (Passive balance)	232 通讯 (232 Communication)	不支持 (Not supported)
UART 接口 (非隔离) UART (non-isolated)	标配 (Standard option)	加热膜功能 (Heating function)	不支持 (Not supported)
开关功能 (Switch function)	选配 (Optional)	蓝牙模块 (Module of Bluetooth)	选配 (Optional)
充电限流功能 (Charging current limit)	选配 (Optional)	电池组串联 (Battery packs in series)	不支持 (Not supported)
电池组并联 (Battery packs in parallel)	不支持 (Not supported)	二次保护功能 (Secondary protection)	不支持 (Not supported)
履历 (History storage)	支持 (Supported)	LCD 显示屏 (LCD display)	选配 (Optional)
预放电功能 (Pre-discharge function)	支持 (Supported)	LED 指示灯接口 (Interface of LED indicator)	不支持 (Not supported)
蜂鸣器 (Buzzer)	选配 (Optional)	GPS 接口 (Interface of GPS)	选配 (Optional)

3.参数设置(Parameter Setting)

3.1.基本参数(Basic parameter)

电芯规格(Cell specifications)	17~24 串铁锂(17~24 strings of LiFePO_4 battery)
接口类型(Interface type)	充放电同口(Charge and discharge are both at the same port)
充电电压(Charging voltage)	3.60V*串数(3.60V*Number of strings)
单体电压范围(Cell voltage range)	2.20~3.75V
持续充电电流(Continuous charging current)	50-200A
持续放电电流(Continuous discharging current)	50-200A
运行功耗(Consumption of running)	$\leq 20\text{mA}$
休眠功耗(Consumption of sleep)	$\leq 300\mu\text{A}$
休眠条件(Sleep conditions)	断开开关或无电流\通讯\保护状态下延时 1min $\pm$ 30S Delay 1min $\pm$ 20s under no current \ communication \ protection state
回路内阻(Circuit resistance)	$\leq 10\text{mR}$
工作温度(Operating temperature)	-20 $^{\circ}\text{C}$ ~75 $^{\circ}\text{C}$
保护板结构尺寸(Structure size of PCB)	
尺寸(size)	170 $\pm$ 2mm * 100 $\pm$ 1mm * 13 $\pm$ 2mm (长度*宽度*高度) (Length*Width*Height)

注：测试需在温度 25 $\pm$ 2 $^{\circ}\text{C}$ ，相对湿度 65 $\pm$ 20% 的环境

Note: Test should be at temperature 25 $\pm$ 2 $^{\circ}\text{C}$, and relative humidity 65 $\pm$ 20% of surroundings.

3.2.主要参数(Main parameter)

		项目(Project)	规格(Specification)			单位 (Unit)
			最小值 MIN	典型值 TYP	最大值 MAX	
过压和欠压保护 (Over-voltage and Under-voltage protection)		过充保护电压(Over-voltage)	3.700	3.750	3.800	V
		过充保护延时(Over-voltage delay)	1000	2000	3000	mS
		过充保护释放(Over-voltage release)	3.55	3.600	3.650	V
		过放保护电压(Under-voltage)	2.100	2.200	2.300	V
		过放保护延时(Under-voltage delay)	1000	2000	3000	mS
		过放保护释放(Under-voltage release)	2.500	2.600	2.700	V
		过放保护释放条件 (Under-voltage release conditions)	电压自恢复或充电恢复 (Self-recovery by increasing voltage or charging)			
		充电过流保护值 (Over-current Charge protection value)	见下面过流保护值配置表 (Refer to configuration table of over-current protection value below)			
		充电过流延时 (Over-current Charge delay)	5	10	15	S
		充电过流释放条件 (Over-current Charge release conditions)	延时 32S 后自动恢复 (Automatic recover after a delay of 32S)			
		一级放电过流保护值 (1th Over-current Discharge value)	见下面过流保护值配置表 (Refer to configuration table of over-current protection value below)			
		一级放电过流保护延迟 (1th Over-current Discharge delay)	5	10	15	S
		二级放电过流保护电流值 (2th Over-current Discharge value)	见下面过流保护值配置表 (Refer to configuration table of over-current protection value below)			
		二级放电过流 2 保护延迟 (2th Over-current Discharge delay)	32	-	500	mS
		放电过流保护恢复条件 (Over-current Discharge release)	延时 32S 后自动恢复 (Automatic recover after a delay of 32S)			
短路保护 (Short Circuit Protection)		短路保护电流 (Short circuit protection current value)	见下面过流保护值配置表 (Refer to configuration table of over-current protection value below)			
		短路保护延迟时间 (Short circuit protection delay time)	62	-	1000	uS
		短路保护恢复 (Short circuit protection recovery)	断开负载后约 5 秒自动释放 Recover by releasing load after approximately 5s			
		短路说明：短路电流小于最小值或高于最大值可能会造成短路保护失效，短路电流超过 2000A ，不保证有短路保护，也不建议做短路保护测试。 (Short-circuit description: The short-circuit current is less than the minimum value or higher than the maximum value, which may cause the short-circuit protection to fail, and the short-circuit current exceeds 2000A , short-circuit protection is not guaranteed, and short-circuit protection testing is not recommended.)				
温度保护 (Short Circuit Discharge)	充电 CHG	温度保护值 (Temperature protection value)	62	65	68	°C
		温度保护释放值 (Temperature protection release value)	52	55	58	°C
		温度保护值 (Temperature protection value)	-13	-10	-7	°C
		温度保护释放值 (Temperature protection release value)	-8	-5	-2	°C
	放电 DSG	温度保护值 (Temperature protection value)	72	75	78	°C
		温度保护释放 (Temperature protection release value)	62	65	68	°C
		温度保护值 (Temperature protection value)	-23	-20	-17	°C
		温度保护释放 (Temperature protection release value)	-13	-10	-7	°C
	FET	温度保护值 (Temperature protection value)	85	90	95	°C
		温度保护释放值 (Temperature protection release value)	65	70	75	°C

均衡功能 (Balance Function)	均衡开启电压 (Balance function turn-on voltage)	3.37	3.40	3.43	V
	开启压差 (Difference opening voltage value)		15		mV
	均衡电流 (Balance current)	20		110	mA
	均衡模式 (Balance model)	静态均衡 (Idle equalization)			
	均衡类型 (Balance type)	脉冲模式 (Pulsed model)			

注：测试需在温度 $25 \pm 2^\circ\text{C}$ ，相对湿度 $65 \pm 20\%$ 的环境。

Note: Test should be at temperature $25 \pm 2^\circ\text{C}$, and relative humidity $65 \pm 20\%$ of surroundings.

过流保护值配置表(Over-current protection value configuration table)

持续电流 (Continuous current)		充电过流保护值 (Charge Over-current value)	一级放电过流保护值 (1 st discharge Over-current value)	二级放电过流保护值 (The second discharge Over-current value)	短路保护值 (Short circuit protection value)
Charge	Discharge				
50A	50A	60 $\pm$ 5A	60 $\pm$ 5A	200 $\pm$ 50A	700 $\pm$ 140A
100A	100A	120 $\pm$ 5A	120 $\pm$ 5A	400 $\pm$ 50A	1600 $\pm$ 400A
150A	150A	170 $\pm$ 5A	170 $\pm$ 5A	500 $\pm$ 50A	2000 $\pm$ 400A
200A	200A	220 $\pm$ 5A	220 $\pm$ 5A	600 $\pm$ 50A	2400 $\pm$ 400A

3.3.参数设置(parameter settings)

JBDTools -V4.2-20220829

通讯口设置 开始 升级

SAVEDATA 保存数据

AFE_SH303 AFE_T1_BQ76XX AFE_DC10XX AFE_O2_3717 AFE_O2_3714 AFE_NUVOTON AFE_SH309

简体中文 中文繁體 English

PASSWORD

电池信息 参数设置 校准 其他功能 历史记录 测试工具

基本保护参数配置

单体过压 3750 mV 释放电压 3600 mV 延时 2 S

单体欠压 2200 mV 释放电压 2600 mV 延时 2 S

整组过压 90000 mV 释放电压 86400 mV 延时 2 S

整组欠压 52800 mV 释放电压 62400 mV 延时 2 S

充电高温 65 C 释放温度 55 C 延时 2 S

充电低温 -10 C 释放温度 -5 C 延时 2 S

放电高温 75 C 释放温度 65 C 延时 2 S

放电低温 -20 C 释放温度 -10 C 延时 2 S

充电过流 120000 mA 释放时间 32 S 延时 10 S

放电过流 120000 mA 释放时间 32 S 延时 10 S

功能配置

SW_EN LOAD_EN BAL_EN CHG_BAL

LED_EN LED_NUM RTC EDV_EN

CHGLimit GPS_EN Buzzer_EN CARMODE

NTC配置

NTC1 NTC2 NTC3 NTC4

NTC5 NTC6 NTC7 NTC8

均衡配置

开启电压 3400 mV GPS关闭电压 2300 mV

均衡精度 15 mV GPS关闭延时 10 S

均衡电流 0 mA

容量配置

标称容量 100000 mAh

循环容量 80000 mAh

单体充满电压 3550 mV

单体截止电压 2500 mV

自放电率 0.1 %

100% 3350 90% 3330

80% 3329 70% 3325

60% 3322 50% 3301

40% 3291 30% 3280

20% 3266 10% 3220

开关 30 S LED 10 S

高级保护

OC2&SC*2

放电过流2值 400 A 过流2延时 32 ms

短路保护值 1600 A 短路延时 500 us

硬件单体过压 3900 mV 过压延时 S

硬件单体欠压 2000 mV 欠压延时 S

短路释放延时 5 S

其他信息配置

检流阻值 0.1 mR 电池串数 24

循环次数 0 序列号 0

制造商 DGJBD

BMS_SN JBD-SP24S004-L24S-100A

生产日期 2022 11 19

Barcode

保护次数

短路次数 0 充电高温 0

充电过流 0 充电低温 0

放电过流 0 放电高温 0

单体过压 0 放电低温 0

单体欠压 0 整体过压 0

RSTNum 0 整体欠压 0

未连接! 串口号:COM11,波特率:9600,数据位:8,校验位:0 参数名称: JBD-SP24S004-L24S-100A.fig 2022-11-28 15:04:56

The diagram of the host computer

***注意事项:**

1.上位机为 **JBDTOOLS - V4.2 或以上版本**，右上角区域选择芯片“**AFE_OZ3714**”。

***Attention:**

1. The version of software in upper computer is **JBDTOOLS - V4.2 or above**, and please choose the 'AFE_OZ3714' at top right corner.

4.功能说明(Function Description)

4.1.过充保护和恢复(Overcharge protection and recovery)

4.1.1.单体过充保护及恢复(Cell overcharge protection and recovery)

当任意一节电芯电压高于单体过充电压设定值，并且持续时间达到单体过充延时，系统进入过充保护状态，关闭充电 MOS，不能对电池充电。

单体过充保护后，当所有单体电压降到单体过充恢复值以下时，解除过充保护状态。也可放电解除。

When the voltage of any cell is higher than the set value of the cell overcharge voltage, and the duration reaches the cell overcharge delay, the system enters the overcharge protection state, the charging MOS will turn off, and the battery cannot be charged.

After the cell overcharge protection, when the voltage of all cells drops below the cell overcharge recovery value, the overcharge protection state is released. It can also be released by discharge.

4.1.2.总体过充保护及恢复(Entire overcharge protection and recovery)

当总体电压高于总体过压设定值，并且持续时间达到总体过充延时，系统进入过充保护状态，关闭充电 MOS，不能对电池充电。

当总体电压降到总电压过压保护恢复值以下时，解除过充保护状态，也可放电解除。

When the entire voltage is higher than the entire Over-voltage set value, and the duration reaches the entire overcharge delay, the system enters the overcharge protection state, turns off the charging MOS, and cannot charge the battery.

When the entire voltage drops below the recovery value of the entire voltage Over-voltage protection, the overcharge protection state is released, and it can also be released by discharge.

4.2.过放保护和恢复(Over-discharge protection and recovery)

4.2.1.单体过放保护及恢复(Cell over-discharge protection and recovery)

当最低节电压低于单体过放电压设定值，并且持续时间达到单体过放延时，系统进入过放保护状态，关闭放电 MOS，不能对电池放电。

发生单体过放保护后，对电池组充电可以解除过放保护状态。

When the minimum cell voltage is lower than the set value of the over-discharge voltage of the cell, and the duration reaches the over-discharge delay of the cell, the system enters the over-discharge protection state, turns off the discharge MOS, and cannot discharge the battery.

After the cell over-discharge protection occurs, charging the battery pack can release the over-discharge protection state.

4.2.2.总体过放保护及恢复(Entire over-discharge protection and recovery)

当总体电压低于总体过放电压设定值，并且持续时间达到总体过放延时，系统进入过放保护状态，关闭放电 MOS，

不能对电池放电。

发生总体过放保护后，对电池组充电可以解除过放保护状态。

When the entire voltage is lower than the entire over-discharge voltage set value, and the duration reaches the entire over-discharge delay, the system enters the over-discharge protection state, turns off the discharge MOS, and cannot discharge the battery.

After the entire over-discharge protection occurs, charging the battery pack can release the over-discharge protection state.

4.3. 充电过流保护和恢复(Over-current protection and recovery in charging)

当充电电流超过充电过流保护电流且持续的时间达到过流检测延迟时间，系统进入到充电过流保护状态，不能对电池进行充电。发生充电过流保护后延时自动恢复，如需不要自动恢复可将对应的释放时间设长；放电也可以解除充电过流状态。

When the charging current exceeds the charging protection current and the duration reaches the Over-current detection delay time, the system enters the charging Over-current protection state and cannot charge the battery. After the charging Over-current protection occurs, it will automatically recover after a delay. If you want to automatically recover or not, you can set the corresponding release time to be longer; the charging Over-current state can also be released by discharging.

4.4. 放电过流保护和恢复(Over-current protection and recovery in discharging)

当放电电流超过放电过流保护电流且持续的时间达到过流检测延迟时间，系统进入到放电过流保护状态，关闭放电 MOS。发生放电过流后延时自动恢复，如需不要自动恢复可将对应的释放时间设长。充电也可以解除放电过流状态。放电有两级过流保护功能，对不同的电流值具有不同的响应速度，更加可靠地保护电池。

When the discharge current exceeds the discharge Over-current protection current and the duration reaches the Over-current detection delay time, the system enters the discharge Over-current protection state and turns off the discharge MOS. Delayed automatic recovery after discharge Over-current occurs, and the corresponding release time can be set longer if automatic recovery is required. Charging can also release the discharge Over-current protect condition. Discharge has two-level Over-current protection function, which has different response speeds for different current values, and protects the battery more reliably.

4.5. 温度保护和恢复(Temperature Protection and Recovery)

4.5.1. 充放电高温保护及恢复(High temperature protection and recovery in charging and discharging)

当充放电时 NTC 检测电芯表面的温度高于设定的高温保护温度时，管理系统进入高温保护状态，充电或放电 MOSFET 关闭，在该状态不能对电池包充电或放电。

当电芯表面的温度下降到高温恢复设定值时，管理系统从高温状态恢复，重新导通充放电 MOS。

When the NTC detects that the temperature of the battery cell surface is higher than the setting of high temperature protection value during charging and discharging, the management system enters the high temperature protection state, the charging or discharging MOSFET is turned off, and the battery pack cannot be charged or discharged in this state.

When the temperature of the surface of the cell drops to the high temperature recovery set value, the management system recovers from the high temperature state and turns on the charge and discharge MOS again.

4.5.2. 充放电低温保护和恢复(Low temperature protection and recovery in charging and discharging)

当充放电时 NTC 检测电芯表面的温度低于设定的低温保护温度时，管理系统进入低温保护状态，充电或放电 MOSFET 关闭，在该状态不能对电池包充电或放电。

当电芯表面的温度上升到低温恢复设定值时，管理系统从低温状态恢复，重新导通充放电 MOS。

When the NTC detects that the temperature of the cell surface is lower than the setting of low temperature protection value during charging and discharging, the management system enters the low temperature protection state, the charging or discharging MOSFET is turned off, and the battery pack cannot be charged or discharged in this state.

When the temperature of the cell surface rises to the low temperature recovery set value, the management system recovers from the low temperature state and turns on the charge and discharge MOS again.

4.6. 均衡功能(Balance function)

管理系统采用电阻旁路的方式进行电芯均衡，充电过程中电池组最高节单体电芯电压达到设定的均衡启动电压值，且电池组单体电芯最低电压与最高电压压差大于设定值时，达到条件的电芯均衡功能开启，相邻的两路均衡不能同时开启。

当电芯压差小于设定值或者电芯电压小于均衡开启电压时均衡停止。

The management system uses the resistance bypass method to balance the cells. During the charging process, the voltage of the highest single cell of the battery pack reaches the set equilibrium starting voltage value, and the voltage difference between the minimum voltage and the maximum voltage of the single cell of the battery pack is greater than the set value. When the value is set, the equalization function of the cells that meet the conditions is enabled, and the two adjacent equalizers cannot be enabled at the same time.

The equalization stops when the cell voltage difference is less than the set value or the cell voltage is less than the equalization turn-on voltage.

4.7. 容量计算(Capacity calculation)

可以通过对电流、时间积分的方式精准地进行电池组的 SOC 计算。电池组满容量、及循环容量可以通过上位机进行设置，在进行完整充放电循环后容量可自动更新。具有充放电循环次数计算功能，当电池组累积放电容量达到设定循环容量时，循环次数增加一次。

The SOC calculation of the battery pack can be accurately performed by integrating current and time. The full capacity and cycle capacity of the battery pack can be set through the host computer, and the capacity can be automatically updated after a complete charge and discharge cycle. It has the function of calculating the number of charge and discharge cycles. When the cumulative discharge capacity of the battery pack reaches the set cycle capacity, the number of cycles increases once.

注：新装电池请根据电池容量设定标称容量和循环容量，并进行一次容量学习，否则可能出现容量不准问题。容量学习操作：先充满电至过压保护，然后放空电至欠压保护，再充一次电即可。

Note: For newly installed batteries, please set the nominal capacity and cycle capacity according to the battery capacity, and conduct a capacity study, otherwise the capacity inaccuracy may occur. Capacity learning operation: first fully charge to Over-voltage protection, then discharge to under-voltage protection, and then charge it again.

4.8. 休眠功能(Sleep function)

保护板处于静态时（无通讯，无电流，无均衡及过压保护），且电压值位于 GPS 关闭电压之上，延时 1-2 分钟后进入休眠状态。而当电压值位于 GPS 关闭电压之下时，保护板会先延时关闭 GPS，之后再进入休眠计时。

进入此状态后，保护板仅降低检测频率和自身功耗。通讯、拨开关、充放电可以自动退出休眠模式。

When the BMS is in static state (no communication, no current, no balance and Over-voltage protection) ,and the voltage value is set above the GPS off voltage.After a delay of 1-2minutes, it will enter the sleep state.However, when voltage value is set below the GPS off voltage, the BMS will delay closing the GPS before entering sleep timing.

After entering this state, the BMS will only reduce the frequency of detecting and its own power consumption.Communication, dial switch, charging and discharging can automatically exit the sleep mode

4.9.通讯功能(Communication)

4.9.1.串口通讯(Serial Communication)



UART 通讯盒
(UART communication box)



RS485 通讯盒
(RS485 communication box)

注：上述工具都需要另行购买。

Note: The above tools need to be purchased separately.

连接方式：在电脑端安装我司通讯盒专用驱动程序后，将通讯盒的 USB 端插在电脑的 USB 端口，另一头接在已经接好电池的保护板对应接口。打开上位机，点通讯口设置，选择通讯盒对应 CMO 口，其他选项不用动，确认后点击开始，即可读取保护内数据。

如需更改保护板参数，一定要先在参数页面点击读取参数后，再来更改参数。

The connection method: After installing the special driver for our communication box on the computer, insert the USB end of the communication box into the USB port of the computer, and connect the other end to the corresponding interface of the protection board that has been connected to the battery. Open the upper computer, click the communication port settings, select the CMO port corresponding to the communication box, and do not change other options. After confirming, click Start to read the data in the protection.

If you need to change the parameters of the protection board, you must click on the parameter page to read the parameters before changing the parameters.

通讯设置：

- 波特率：9600；
- 校验位：无；
- 数据位长度：8 位；

- 停止位长度：1 位

COM Settings:

- Baud rate:9600;
- Parity Bit:NONE;
- Data Bit:8 bits;
- Stop Bit:1 bit

4.9.2.CAN 总线(CAN BUS)

环境配置：将 USBCAN Driver 安装至电脑中，首先需要核对电脑操作系统，32 位操作系统与 64 位操作系统匹配不同的驱动文件。（32 位操作系统匹配文件后缀“x86”，64 位操作系统匹配文件后缀“x64”）。最后可在电脑的设备管理器中查看端口以检查是否成功安装。

连接方式：将通讯盒的 USB 线插入电脑的 USB 端口，另一端连接在电池的保护板对应接口。

通讯格式：ID 默认状态下选择 CAN_ID_0，CAN 设备根据通讯盒类型选择，波特率默认为 500K，通道选择默认为 0。

Environment configuration: To install the USBCAN Driver into computer, you need to check the computer operating system initially. The 32-bit operating system and the 64 bit operating system match different driver files. (32-bit operating system matching file suffix "x86", 64 bit operating system matching file suffix "x64"). Finally, find the port in the device manager of the computer to check whether the installation is successful.

Connection method: Insert the USB cable of the communication box into the USB port of the computer, and connect the other end to the corresponding interface of the BMS.

Connection format: ID Address is CAN_ID_0 by default. The can device is selected according to the type of communication box. The baud rate is 500K by default, and the channel selection is 0 by default.

5.主要物料(Main material)

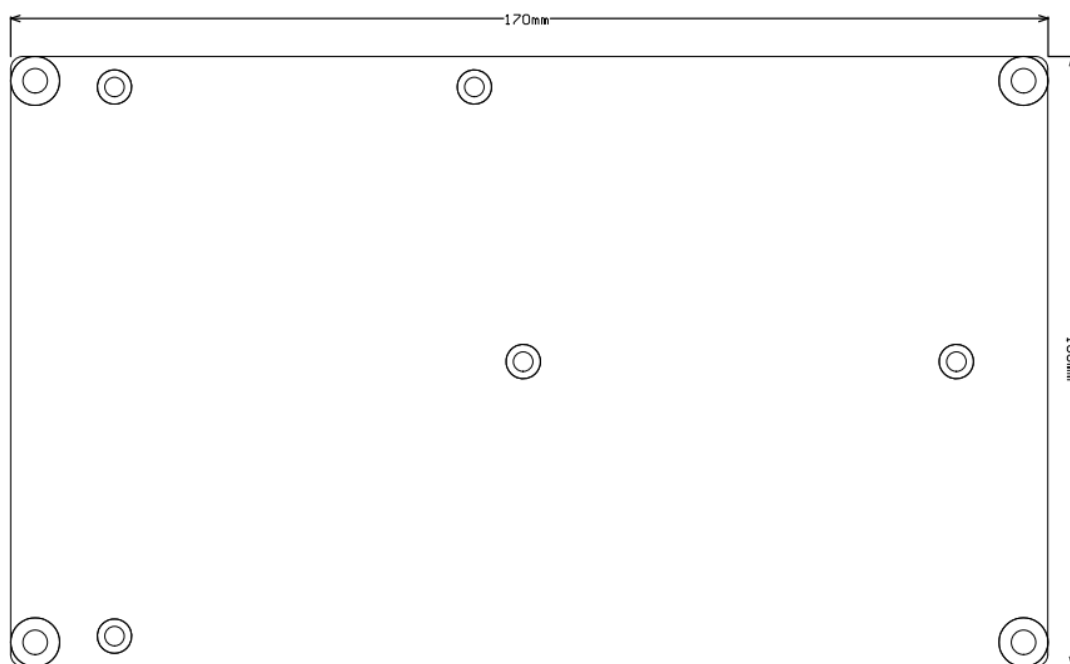
序号 (Number)	物料名称 (Name of Material)	生产厂家 (Manufacturer)	数量 (Quantity)
1	IC\QFP\OZ3714\QFN32L	O ₂ Micro(凹凸)	2PCS
2	IC\QFP\HC32L072KATA\QFP64	HUADA(华大)	1PCS
3	MOS\SMD\G017N10NS1TA\TOLL	HUAYI-Micro(华羿微)	XPCS
4	NTC*2\10K\3950\250mm\带端子\HY2.0	仙桥	2PCS
配件(Accessories)			
1	采集线\15PIN\HY2.0\带卡扣\24AWG\550mm\黑白红	---	1PCS
2	采集线\10PIN\HY2.0\带卡扣\24AWG\550mm\黑白红	---	1PCS
3	采集线\2PIN\HY2.0\带卡扣\24AWG\550mm\蓝白	---	1PCS

注：以上物料可能用同等规格参数或者更好的规格参数的物料替代，如有认证需求不允许更换物料，需要通知我司业务重新送样，受控规格书，最终解释权归嘉佰达所有。

Note: The above materials may be replaced by materials with the same specifications or better specifications. If there are certification requirements, the replacement of materials is not allowed, and we need to notify our business to send samples again. The controlled specifications, the final interpretation right belongs to JBD.

6.示意图及尺寸(Schematic and Dimensions)

6.1.尺寸及安装点标注图(Dimensions and installation point drawing)

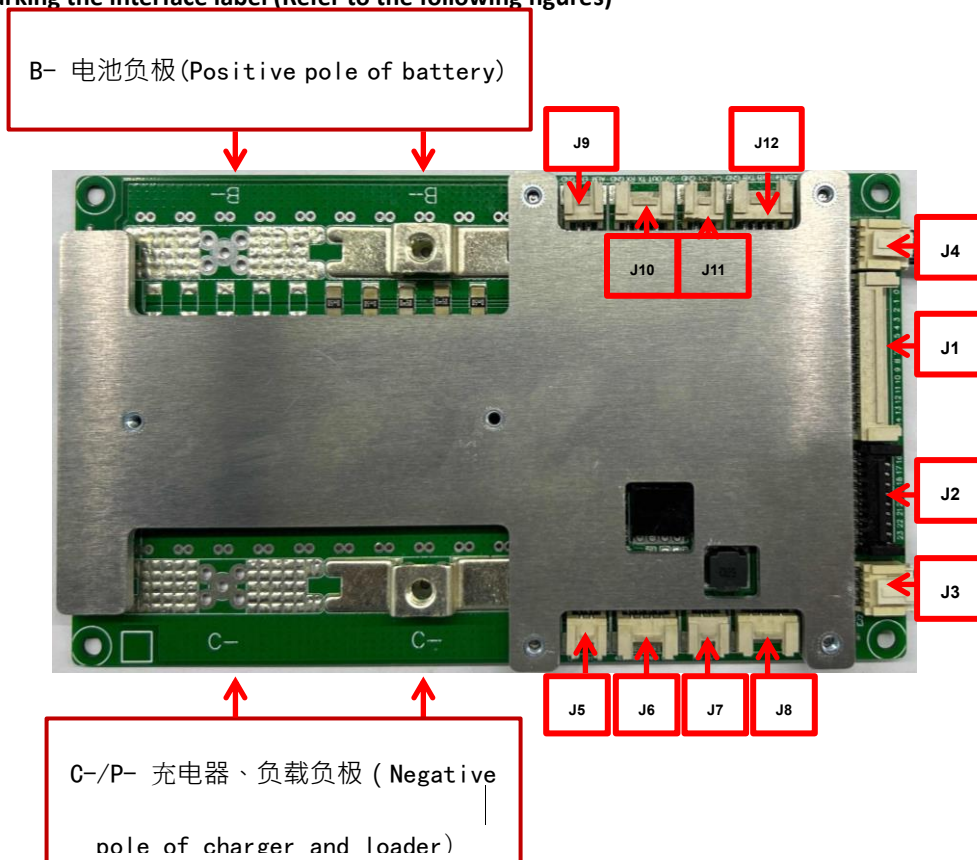


Structure size

7.信号口定义(Definition of signal port)

7.1.示意图标注接口标号(Illustration annotation)

Schematic marking the interface label (Refer to the following figures)



标号 (Label)	位号 (Tag number)	接插件功能 (Connector)	接插件示意图 (Schematic diagram)	PIN	PIN 功能定义 (Pin function definition)	备注 (Note)
1	J1 (HY2.0-15P) (带卡扣)	电压检测插座 (Voltage detection socket)		1	接最低节电芯负极 Connect to Negative Side of Cell 1	B-
				2	接第 1 节电芯正极 Connect to Positive Side of Cell 1	
				3	接第 2 节电芯正极 Connect to Positive Side of Cell 2	
				4	接第 3 节电芯正极 Connect to Positive Side of Cell 3	
				5	接第 4 节电芯正极 Connect to Positive Side of Cell 4	
				6	接第 5 节电芯正极 Connect to Positive Side of Cell 5	
				7	接第 6 节电芯正极 Connect to Positive Side of Cell 6	
				8	接第 7 节电芯正极 Connect to Positive Side of Cell 7	
				9	接第 8 节电芯正极 Connect to Positive Side of Cell 8	
				10	接第 9 节电芯正极 Connect to Positive Side of Cell 9	
				11	接第 10 节电芯正极 Connect to Positive Side of Cell 10	
				12	接第 11 节电芯正极 Connect to Positive Side of Cell 11	
				13	接第 12 节电芯正极 Connect to Positive Side of Cell 12	
				14	接第 13 节电芯正极 Connect to Positive Side of Cell 13	
				15	接第 14 节电芯正极 Connect to Positive Side of Cell 14	

2	J2 (HY2.0-10P) (带卡扣)	电压检测插座 (Voltage detection socket)		1	接第 15 节电芯正极 Connect to Positive Side of Cell 15	
				2	接第 16 节电芯正极 Connect to Positive Side of Cell 16	
				3	接第 17 节电芯正极 Connect to Positive Side of Cell 17	
				4	接第 18 节电芯正极 Connect to Positive Side of Cell 18	
				5	接第 19 节电芯正极 Connect to Positive Side of Cell 19	
				6	接第 20 节电芯正极 Connect to Positive Side of Cell 20	
				7	接第 21 节电芯正极 Connect to Positive Side of Cell 21	
				8	接第 22 节电芯正极 Connect to Positive Side of Cell 22	
				9	接第 23 节电芯正极 Connect to Positive Side of Cell 23	
				10	接第 24 节电芯正极 Connect to Positive Side of Cell 24	
3	J3 (HY2.0-4P) (带卡扣)	NTC		1	连接温度探头 Connect to the temperature detection	
				2		
				3		
				4		
4	J4 (HY2.0-4P) (带卡扣)	NTC		1	连接温度探头 Connect to the temperature detection	
				2		
				3		
				4		
5	J5 (HY2.0-2P) (带卡扣)	充电器检测端口 (Charger detection port)		1	CH-	R-ID
				2	CH_ID	Chrger-
6	J6 (HY2.0-4P) (带卡扣)	通讯接口 (Communication port)		1	CAN - Low	CAN-BUS
				2	CAN - High	
				3	RS485 - B	RS485
				4	RS485 - A	
7	J7 (HY2.0-2P) (带卡扣)	弱电开关 (Switch of discharging)		1	K -	
				2	K +	
8	J8 (HY2.0-4P) (带卡扣)	UART 通讯接口 (非隔离) UART (Non-Isolated)		1	UART - GND	蓝牙接口
				2	UART - RXD	
				3	UART - TXD	
				4	UART - VDD 11-14V	
9	J9 (HY2.0-2P) (带卡扣)	报警驱动 (Alarm)		1	ALARM - GND	
				2	ALARM - EN	14V
10	J10 (HY2.0-4P) (带卡扣)	UART 通讯接口 (非隔离) UART (Non-Isolated)		1	UART - GND	RFID 接口
				2	UART - RXD	
				3	UART - TXD	
				4	UART - VDD 5V	

11	J11 (HY2.0-2P) (带卡扣)	限流模块驱动		1	CM - GND	
				2	CM - EN	
12	J12 (HY2.0-4P) (带卡扣)	RS485 可配置为 GPS 或显示屏 接口 (更改硬件)		1	GND	
				2	RS485 - A	不隔离
				3	RS485 - B	
				4	B+	

7.2.BMS 电池串数连接(BMS Cell Connections)

Cell connection for SP24S004

	24S	23S	22S	21S	20S	19S	18S	17S	16S	15S	14S
BC24	CELL24+	CELL23+	CELL22+	CELL21+	CELL20+	CELL19+	CELL18+	CELL17+	CELL16+	CELL15+	短接到 BC14
BC23	CELL23+	CELL22+	CELL21+	CELL20+	CELL19+	CELL18+	CELL17+	CELL16+			
BC22	CELL22+	CELL21+	CELL20+	CELL19+	CELL18+	CELL17+	CELL16+	CELL15+			
BC21	CELL21+	CELL20+	CELL19+	CELL18+	CELL17+	CELL16+	CELL15+	CELL14+	CELL15+	CELL14+	
BC20	CELL20+	CELL19+	CELL18+	CELL17+	CELL16+	CELL15+	CELL14+	CELL13+	CELL14+	CELL13+	
BC19	CELL19+	CELL18+	CELL17+	CELL16+	CELL15+	CELL14+	CELL13+	CELL12+	CELL13+	CELL12+	
BC18	CELL18+	CELL17+	CELL16+	CELL15+	CELL14+	CELL13+	CELL12+	CELL11+	CELL12+	CELL11+	
BC17	CELL17+	CELL16+	CELL15+	CELL14+	CELL13+	CELL12+	CELL11+	CELL10+	CELL11+	CELL10+	
BC16	CELL16+	CELL15+	CELL14+	CELL13+	CELL12+	CELL11+	CELL10+	CELL9+	CELL10+	CELL9+	
BC15	CELL15+	CELL14+	CELL13+	CELL12+	CELL11+	CELL10+	CELL9+	CELL8+	CELL9+	CELL8+	
BC14	CELL14+	CELL13+	CELL12+	CELL11+	CELL10+	CELL9+	CELL8+	CELL7+	CELL8+	CELL7+	CELL14+
BC13	CELL13+										CELL13+
BC12	CELL12+	CELL12+									CELL12+
BC11	CELL11+	CELL11+	CELL11+								CELL11+
BC10	CELL10+	CELL10+	CELL10+	CELL10+							CELL10+
BC9	CELL9+	CELL9+	CELL9+	CELL9+	CELL9+						CELL9+
BC8	CELL8+	CELL8+	CELL8+	CELL8+	CELL8+	CELL8+					CELL8+
BC7	CELL7+	CELL7+	CELL7+	CELL7+	CELL7+	CELL7+	CELL7+		CELL7+		CELL7+
BC6	CELL6+	CELL6+	CELL6+	CELL6+	CELL6+	CELL6+	CELL6+	CELL6+	CELL6+	CELL6+	CELL6+
BC5	CELL5+	CELL5+	CELL5+	CELL5+	CELL5+	CELL5+	CELL5+	CELL5+	CELL5+	CELL5+	CELL5+
BC4	CELL4+	CELL4+	CELL4+	CELL4+	CELL4+	CELL4+	CELL4+	CELL4+	CELL4+	CELL4+	CELL4+
BC3	CELL3+	CELL3+	CELL3+	CELL3+	CELL3+	CELL3+	CELL3+	CELL3+	CELL3+	CELL3+	CELL3+
BC2	CELL2+	CELL2+	CELL2+	CELL2+	CELL2+	CELL2+	CELL2+	CELL2+	CELL2+	CELL2+	CELL2+
BC1	CELL1+	CELL1+	CELL1+	CELL1+	CELL1+	CELL1+	CELL1+	CELL1+	CELL1+	CELL1+	CELL1+
BC0	CELL1-	CELL1-	CELL1-	CELL1-	CELL1-	CELL1-	CELL1-	CELL1-	CELL1-	CELL1-	CELL1-

8.环境适用性(Environmental suitability)

8.1.工作环境(The environment of working)

●BMS 保护板允许在下列条件下正常工作：

●环境温度：-20°C ~ 75°C；

●相对湿度：5% ~ 90%；

●大气压力：86kPa~106 kPa；

●BMS The protective plate allows normal operation under the following conditions:

●Ambient temperature: -20°C ~ 75°C;

●Relative humidity: 5% ~ 90%;

●Atmospheric pressure: 86kPa~106kPa;

8.2.存储环境(The environment of storage)

●BMS 保护板应存储在环境温度为-5° C~+40° C、相对湿度不大于 70%、清洁通风良好的库房内，空气中不得含有腐蚀性气体及影响电气绝缘的介质，不得受任何机械冲击或重压。不受阳光直射，与热源(暖气设备等)之间的距离不得少于 2m。在以上存储条件下，BMS 保护板可存放一年。

●BMS should be stored in a clean and well-ventilated warehouse with an ambient temperature of -5°C~+40°C, a relative humidity of not more than 70%, and the air must not contain corrosive gases and media that affect electrical insulation, and must not be affected by any mechanical Shock or heavy pressure. Not subject to direct sunlight, and the distance from the heat source (heating equipment, etc.) should not be less than 2m. Under the above storage conditions, the BMS protection board can be stored for one year.

9. 包装运输(Packing and shipping)

9.1. 标志(Logo)

BMS 保护板应有下列清晰耐久标志：

- 产品名称、型号
- 电芯型号
- 出厂日期及编号

9.2. 包装(Package)

- 包装应符合防潮、防振动的要求，包装箱应牢固可靠，箱内应衬有防潮材料，产品在箱内不应窜动。
- 外部纸箱包装箱，单板防静电袋加气泡袋包装；

● The packaging should meet the requirements of moisture-proof and anti-vibration, the packing box should be firm and reliable, the inside of the box should be lined with moisture-proof material, and the product should not move in the box.

- External carton box, veneer anti-static bag plus bubble bag packaging;

9.3. 运输(transportation)

● 在运输中，产品不得受剧烈机械冲撞、暴晒、雨淋、化学腐蚀性物品及有害气体侵蚀；在装卸过程中，产品轻搬轻放，严禁摔掷、重压。

- 包装箱码放高度小于 5 层。

● During transportation, the product shall not be subject to severe mechanical impact, exposure to the sun, rain, chemical corrosive substances and harmful gases; During the loading and unloading process, the product should be handled with care, and it is strictly forbidden to throw or press it.

- The height of the packing boxes shall be less than 5 layers.

10.注意事项(Precautions)

1. 本电池管理系统常规是不能串联使用的，需要定制版本才支持串联使用。
2. 多个使用本管理系统的电池包并联时，应确保并联之前各电池包的最大压差低于 3V。
3. 多个使用本管理系统的电池包并联使用时，适配器总的充电冲击电流可能施加到单个电池包上，应确保适配器总的充电冲击电流不超过单个管理系统充电冲击电流的最大值。
4. 本管理系统的短路保护功能适用于多种应用情景，但并不能保证可以在任意条件下短路。当电池包和短路回路的内阻值总和低于 40mΩ、电池组容量超出额定值 20%、短路电流超过 2000A、短路回路的电感非常大或者短路的导线总长度非常长时，请自行测试确定是否可以使用本管理系统。
5. 焊接电池引线时，一定不可有错接或反接。如果确实已接错，这块电路板可能已损坏，需要重新测试合格后才可使用。
6. 装配时管理系统不要直接接触到电芯表面，以免损坏电路板。装配要牢固可靠。
7. 使用中注意引线头、烙铁、焊锡等不要碰到电路板上的元器件，否则有可能损坏本电路板。焊接本电路板请不要使用膏状助焊剂，否则有可能导致本电路板工作不正常。
8. 使用过程要注意防静电、防潮、防水等。
9. 使用过程中请遵循设计参数及使用条件，不得超过本规格书中的值，否则有可能损坏管理系统。
10. 将电池组和管理系统组合好以后，初次上电如发现无电压输出或充不进电，请检查接线是否正确。
11. 本规格书中的参数、功能和外形仅供参考，请以保护板实物为准。

1.This battery management system cannot be used in series generally, and requires a customized version to support series use.

2.When multiple battery packs using this management system are connected in parallel, make sure that the maximum voltage difference of each battery pack is lower than 3V before parallel connection.

3.When multiple battery packs using this management system are used in parallel, the total charging inrush current of the adapter may be applied to a single battery pack. It should be ensured that the total charging inrush current of the adapter does not exceed the maximum charging inrush current of a single management system.

4.The short-circuit protection function of this management system is suitable for a variety of application scenarios, but it does not guarantee that it can be short-circuited under any conditions. When the total internal resistance of the battery pack and the short-circuit loop is lower than 40mΩ, the capacity of the battery pack exceeds the rated value by 20%, the short-circuit current exceeds 2000A, the inductance of the short-circuit loop is very large, or the total length of the short-circuit wire is very long, please test yourself to determine whether This management system can be used.

5.When soldering the battery leads, there must be no wrong or reverse connection. If it is indeed connected incorrectly, the circuit board may be damaged and needs to be re-tested before it can be used.

6.When assembling, the management system should not directly touch the surface of the cell to avoid damage to the circuit board. Assembly should be firm and reliable.

7.During use, be careful not to touch the components on the circuit board such as lead tips, soldering iron, solder, etc., otherwise the circuit board may be damaged. Please do not use paste flux when soldering this circuit board, otherwise it may cause this circuit board to work abnormally.

8.During use, pay attention to anti-static, moisture-proof, waterproof, etc.

9.During use, please follow the design parameters and conditions of use, and must not exceed the values in this specification, otherwise the management system may be damaged.

10.After the battery pack and the management system are combined, please check whether the wiring is correct if you find that there is no voltage output or charging fails when the battery is powered on for the first time.

11.The Parameter,function and outlook of BMS in this specification are for reference only,please refer to actual product.